Khulna University of Engineering & Technology

Department of Electronics and Communication Engineering

ECE 2200: Electronic Circuits Design Laboratory | Open-Ended Project

Wireless Power Transfer

Short-Range Inductive Mobile Charging System

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Objectives

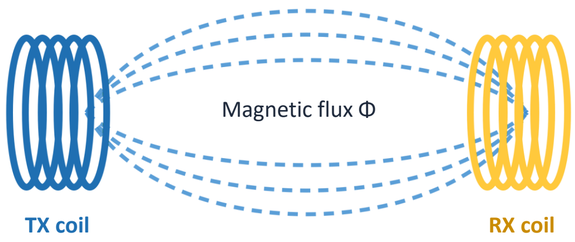
- Design and build a short-range inductive wireless power transfer system
- Study electromagnetic induction and calculate circuit parameters
- Measure voltage, current and efficiency against spacing, alignment, orientation, frequency and load

Contributions

- PWM push-pull transmitter (KA3525A, 2× IRFZ44N)
- Wound resonant TX and RX coils
- 5 V regulated receiver with USB Type-C output
- Wireless transfer across ≈2.5 cm gap
- Breadboard design moved to a PCB

Background & Motivation

- Connector-free charging for phones and portable devices
- Energy crosses a short air gap by magnetic coupling
- Combines PWM, MOSFET switching, resonance, rectification and regulation
- Coupling drops with distance and misalignment
- Low-power educational prototype, not a certified fast charger



Faraday's law

$$\varepsilon = -N\frac{d\Phi}{dt}$$

LC resonance

$$f = \frac{1}{2\pi\sqrt{LC}}$$

Project Methodology

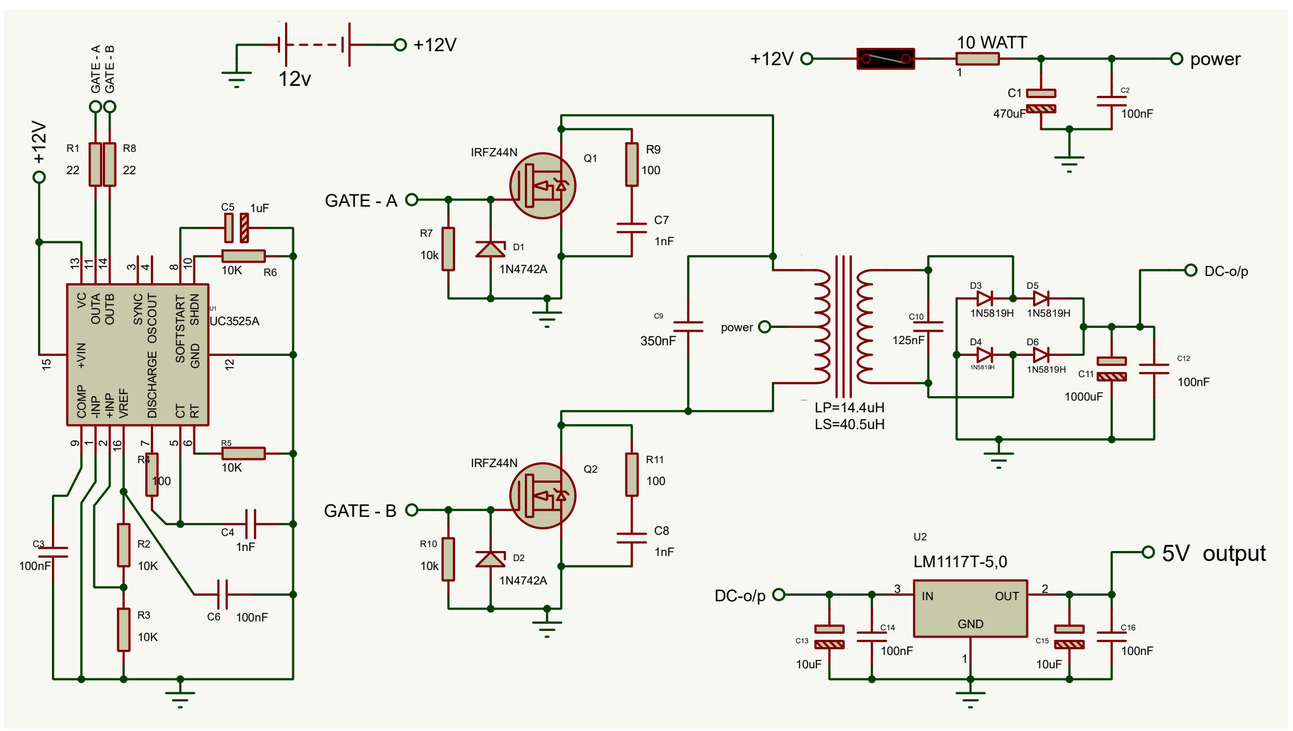
TRANSMITTER

12 V DC supply (fuse + filter) → KA3525A PWM (≈75 kHz drive) → Gate drive (22 Ω + zener) → 2× IRFZ44N (push-pull switch) → TX coil (6-0-6 turns, 300 nF)

MAGNETIC COUPLING · 2–5 mm gap

RECEIVER

RX coil (18–20 turns, 100 nF) → Schottky bridge (4× 1N5819) → Filter caps (1000 µF + 100 nF) → 7805 regulator (≈5 V DC) → USB Type-C (5 V output)

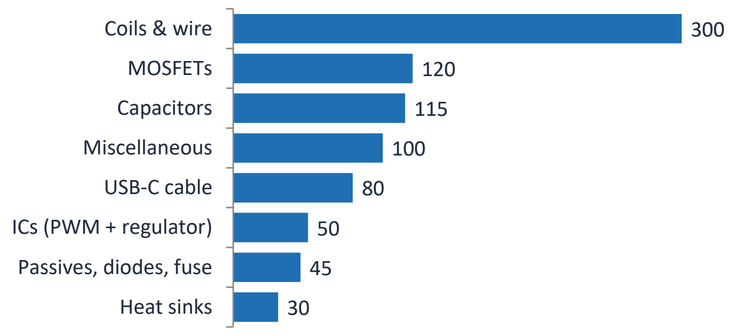


Circuit diagram (Proteus): KA3525A driver, IRFZ44N push-pull stage, rectifier and 5 V regulator

Apparatus & Budget

BDT 840 estimated total cost

Cost by category (BDT)



Component	Qty	BDT
KA3525A PWM controller	1	35
IRFZ44N MOSFET	2	120
TX / RX coils (5.5 m wire)	2	300
Polypropylene capacitors	4	40
1N5819 Schottky diodes	4	10
7805 regulator	1	15
Electrolytic & ceramic caps	15	75
Resistors	11	10
Zener diodes (15 V)	2	10
Fuse (3 A)	1	5
Current-limiting resistor	1	10
USB Type-C cable	1	80
Heat sink	3	30
Miscellaneous materials	—	100
Total		**840**

SOFTWARE & TEST EQUIPMENT

- Proteus simulation
- Digital multimeter
- Oscilloscope
- Load resistors

Complex Engineering

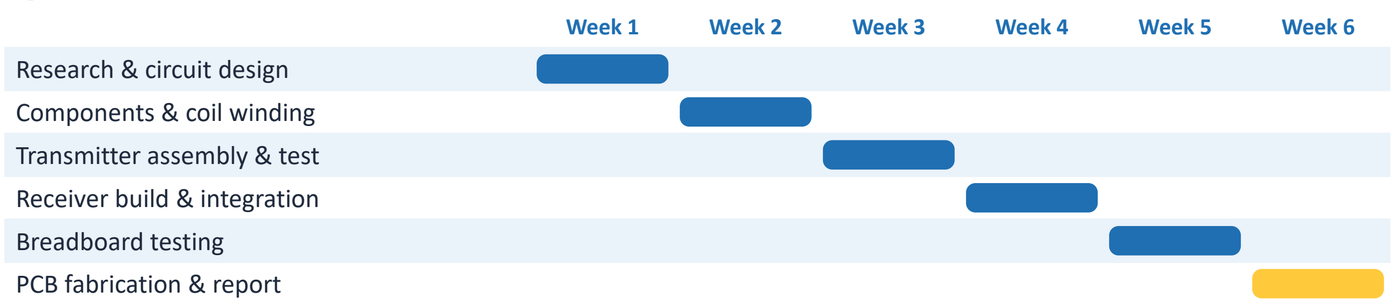
Problems addressed

- **P1 Depth of knowledge:** PWM, switching, resonance, induction
- **P2 Conflicting needs:** distance, power, frequency, heat
- **P3 Depth of analysis:** coil L, resonance, losses, load
- **P4 Familiar issues:** parasitics, coupling, alignment
- **P7 Interdependence:** frequency, coils, rectifier, regulator

Activities addressed

- **A1 Range of resources:** PWM IC, MOSFETs, coils, Proteus, instruments
- **A2 Level of interaction:** driver, gate drive, coils, rectifier
- **A4 Society & environment:** low-power, protected design
- **A5 Familiarity:** standard blocks, new high-frequency behaviour

Work Timeline



Future Aspects

- Measure power, efficiency and temperature
- Optimize coil design for longer range
- Improve alignment tolerance and coupling
- Replace the 7805 with a switching regulator
- Reduce switching, coil and rectifier losses
- Test stability under varied loads